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Global, regional, and national economic value of reducing amenable tuberculosis mortality

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A B S T R A C T

Background: Progress in reducing Tuberculosis (TB) mortality falls short of one-third of the End TB Strategy target, partly due to a high proportion of amenable deaths, most of which occur among individuals aged 15–64 years. The loss of this working-age population has a deterrent effect on human capital and economic growth. Our objective was to estimate the global, regional, and national economic value of reducing amenable TB mortality at the macroeconomic level.

Methods: We utilized TB burden data from the 2021 Global Burden of Disease (GBD) to estimate amenable TB mortality across countries and regions, stratified by sex, age, and drug-resistance type. Economic data were obtained from the World Bank. We estimated the economic losses due to amenable TB mortality using the Value of Lost Welfare (VLW) approach, which integrates the Value of Statistical Life (VSL), Years of Life Lost (YLL) and Gross Domestic Product (GDP). The age-specific percentage of economic losses relative to GDP was calculated.

Results: The global number of amenable TB deaths in 2021 was 616,736, accounting for 53.04% of total TB mortality. The global economic losses due to amenable TB mortality in 2021 were estimated at $480.75 billion, accounting for 0.31% of global GDP. Losses among individuals aged 15–49 and 50–64 accounted for 48.73% and 34.12%, respectively., respectively. Lower-middle-income countries had the largest economic losses at $376.77 billion, corresponding to 1.30% of their GDP, while the percentage of economic losses relative to GDP was highest in low-income countries at 1.73%, with losses of $24.79 billion.

Conclusion: Amenable TB mortality can lead to disproportionate economic losses, especially among deaths in people aged 15-49 years and in lower-middle-income countries. As these deaths and economic losses can be avoidable, investments in improving healthcare services will bring huge economic value.

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1. Introduction

Although Tuberculosis (TB) is preventable and largely curable, TB remains the leading cause of death as a single infectious agent globally, with 10.8 million incident cases and 1.3 million deaths annually worldwide in 2023 (WHO, 2024). Global initiatives have been performed prominently and achieved progress in reducing TB incidence and mortality rates by 8.3% and 23% between 2015 and 2023, respectively (WHO, 2024). However, these reductions fall far short of the "End TB Strategy" targets of a 50% reduction in incidence and 75% reduction in mortality by 2025 (WHO, 2024). One potential challenge to achieve the goal of decreasing mortality is that about 55% of amenable TB deaths have not been avoided due to weak health systems, which result in low-quality health care and poor accessibility (Floyd et al., 2018; Kruk et al., 2018; Wolde et al., 2025). Previous studies have shown that investments in health systems can decrease amenable mortality (Jesus et al., 2025; Mackenbach et al., 2017; Maruthappu et al., 2016), but the economic value of reducing amenable TB mortality is unknown.

Numerous studies have explored the economic impact of TB at the microeconomic level, evaluating whether TB services or technologies were cost-effective from a healthcare provider and societal perspective (Brümmer et al., 2024; Menzies et al., 2016; Portnoy, Clark, et al., 2023; Ryckman et al., 2023, 2024), or alternatively, estimating the national economic burden by scaling individual and household costs (Portnoy, Yamanaka, et al., 2023; Su et al., 2020). However, more than 70% of deaths occurred among people aged 15-64 (Collaborators, 2024a), and the loss of working-age people implies a reduction in human capital, which has a deteremental effect on economic growth (Laxminarayan, 2007). Therefore, macroeconomic study that can capture the effect of improved health on national economic growth are needed. Ranganathan et al. estimated the welfare lost due to TB-related DALYs in South Asia and Sub-Saharan Africa in 2015 (Ranganathan et al., 2021). Silva et al. (Silva et al., 2021) estimated the full-income loss due to TB mortality in 2018 for 120 countries and further explored the economic consequences of not meeting the target of reducing TB mortality by 90% in 2030. However, the economic value of reducing amenable TB mortality that could be prevented through timely and effective medical interventions (GHAaQ, 2017) remains unclear at the macroeconomic level. Estimating this economic value can bridge the gap in increasing health investments between ministries of health and finance, which has shown promising application in resource allocation for stroke (Gerstl et al., 2023), and some noncommunicable diseases and injuries (Chang et al., 2024; Verguet et al., 2024).

In this paper, we used a macroeconomic model to assess the economic value of reducing amenable TB mortality at the global, regional, and national levels based on the 2021 Global Burden of Disease (GBD) database and the World Development Indicators database. We estimated the economic losses associated with amenable TB mortality using the value of lost welfare (VLW) approach, which incorporates the concepts of the value of statistical life (VSL) and years of life lost (YLL) (Alkire et al., 2018; Jamison et al., 2013). Our results will provide important insights into the potential benefits of increasing health investments in TB control.

2. Methods

2.1. Data sources

We extracted data from the Global Health Data Exchange (GHDx) tool (https://ghdx.healthdata.org) based on the GBD 2021 study, which evaluated the global burden of diseases, injuries, and risk factors across different age and sex groups (GDaI, 2024). The GBD data included age- and sex-specific estimates for total mortality, incidence and mortality attributed to risk factors for HIV-negative TB, drug-susceptible TB (DS-TB), multidrug-resistant TB (MDR-TB), and extensively drug-resistant TB (XDR-TB) in 203 countries (Collaborators, 2024a). Additionally, we gathered data on population, life expectancy, and the Socio-Demographic Index (SDI) for each country in 2021 from the GBD study (Global age, 2024). Economic data on gross domestic product (GDP), GDP per capita, and current health expenditure per capita for 186 countries in 2021 were obtained from the World Development Indicators Database provided by the World Bank (Bank). All costs are presented in 2021 international dollars, adjusted for purchasing power parity (PPP).

2.2. Estimating amenable TB mortality

Based on the definition of amenable mortality in previous studies (Alkire et al., 2018; GHAaQ, 2017), we decomposed TB mortality into three mutually exclusive components: preventable, unavoidable, and amenable mortality. Preventable TB mortality includes deaths that could be averted through public health interventions aimed at modifying risk factors (Alkire et al., 2018). Unavoidable TB mortality was determined by assuming that the lowest case-fatality rate (CFR) observed across the 21 GBD regions represents the best-case scenario for healthcare quality and access to health services (Alkire et al., 2018).

Amenable mortality was defined as the difference between the remaining mortality (after total mortality subtracting preventable mortality) and unavoidable mortality. Individuals aged 75 years and older are excluded from amenable mortality analysis as standard practices (Development OfEC-oa, 2022).

Preventable TB mortality (PM) was derived from the GBD 2021 comparative risk assessment estimates. In GBD 2021, risk factors are classified into behavioral, environmental/occupational, and metabolic categories. For HIV-negative TB mortality, attributable risks primarily include behavioral and metabolic factors (Collaborators, 2024b). Because our objective was to quantify mortality potentially reducible through healthcare service improvements, we defined preventable mortality as the portion of HIV-negative TB deaths attributable to behavioral risks only. We explicitly excluded deaths attributable to metabolic risks (e.g., high fasting plasma glucose and high body-mass index), as these can often be modified through clinical management and conceptually overlap with mortality amenable to healthcare (Alkire et al., 2018). Behavioral risks for TB in the GBD framework include tobacco use, alcohol use, dietary risks, and low physical activity. We therefore extracted GBD 2021 estimates of HIV-negative TB mortality attributable to these behavioral risk factors to represent preventable TB mortality. Globally, these risks accounted for 19.43% of TB deaths in 2021 (Collaborators, 2024b).

For each age group *a*, sex *s*, country *c*, and drug resistance type *r*, we defined remaining mortality (*RM*) as total TB mortality minus preventable mortality:

$$RM_{a,s,c,r} = M_{a,s,c,r} - PM_{a,s,c,r} \tag{1}$$

where $M_{a,s,c,r}$ is the total TB mortality and $PM_{a,s,c,r}$ is the preventable TB mortality.

To estimate the unavoidable mortality (*UM*), we first calculated the CFR for each sex, age group, and drug-resistance type across 21 regions (*i*) as defined by the GBD study. We assumed that the minimum regional CFR (*mCFR*) represents the best-case scenario under optimal healthcare accessibility (the lowest unavoidable case fatality rate) (Alkire et al., 2018). The *mCFR* was estimated at the regional level to mitigate the influence of extreme values from countries with small populations. The GBD provides estimates of TB incidence ($I_{a,s,r}$) (Collaborators, 2024a). The minimum regional CFR for each age group, sex, and drug-resistance type combination is defined by the following formula (Alkire et al., 2018):

$$mCFR_{a,s,r} = \min\left[CFR_{a,s,i,r}\right] = \min\left[\frac{RM_{a,s,i,r}}{I_{a,s,i,r}}\right] \tag{2}$$

If *mCFR* represents the best attainable CFR, then unavoidable mortality at the country level is defined by:

$$UM_{a,s,c,r} = mCFR_{a,s,r} \times I_{a,s,c,r} \tag{3}$$

Finally, amenable mortality (*AM*) is estimated with the following:

$$AM_{a,s,c,r} = max\left[RM_{a,s,c,r} - UM_{a,s,c,r}, 0\right] \tag{4}$$

For each country, we estimated the amenable TB mortality stratified by age, sex, and drug resistance type in 2021. The total amenable TB mortality for each country and regions was derived by summing the amenable mortality across all age-sex-drug resistance subsets. Additionally, we calculated the percentage of amenable TB mortality relative to total TB mortality (Fig.1, S1). At the regional (global) level, the amenable TB mortality rate and its percentage were calculated by aggregating the values from the countries in the region (or all 203 countries) (Table 1, Table S1, Fig. 1). We also calculated these metrics for different income groups (low-income, lower-middle-income, upper-middle-income, and high-income) (BANK, 2025) (Table S2).

2.3. Economic value of reducing amenable TB mortality

We estimated the economic value of reducing amenable TB mortality using Value of Lost Welfare (VLW) method, a standardized framework that quantifies the economic burden of premature mortality by integrating Years of Life Lost (YLLs) and the Value of a Statistical Life (VSL). (Gerstl et al., 2023).

To account for the time value of money, we applied a 3% annual discount rate, consistent with standard practices in health economic evaluations (Chudasama et al., 2022). The inclusion of discounting results in the following formula for calculating YLL per death (Ranganathan et al., 2021):

$$YLL_{a,c,s} = \int_a^{LE_{a,c,s}} e^{-dr(x-a)}dx \tag{5}$$

where $a = age\ of\ death$, $LE_{a,c,s} =$ age, country and sex specific life expectancy in 2021, $x = age\ integrated\ over\ YLLs$, $dr = discount\ rate$.

We then calculated the total YLLs attributable to amenable TB mortality as:

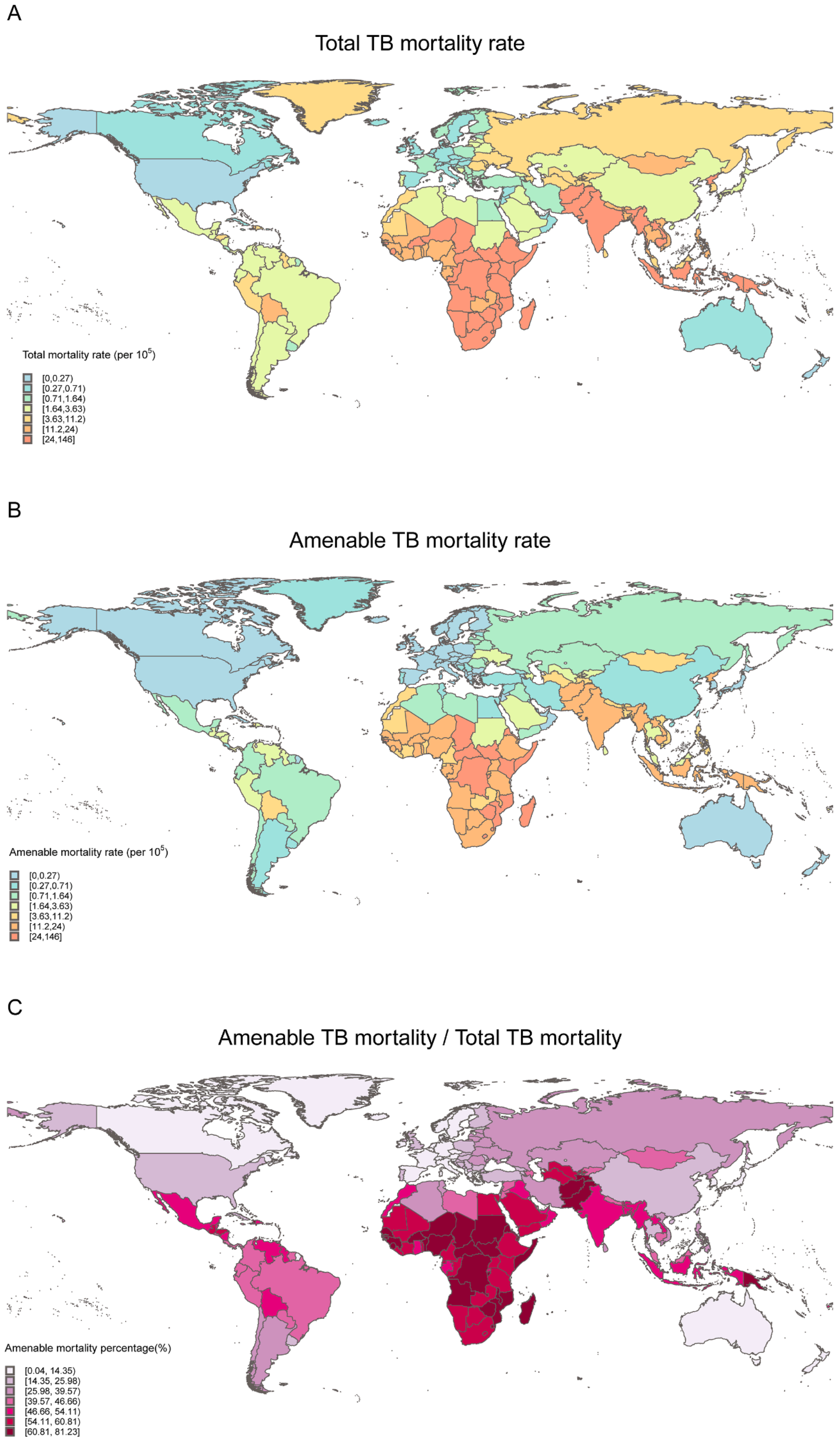

Fig. 1. HIV-negative TB (A) total mortality rates, (B) amenable mortality rates, (C) the percentage of mortality is amenable in 2021 in 203 countries and territories.

$$YLL_{AM(a,c,s)} = YLL_{a,c,s} \times AM_{a,c,s} \times Population_{a,c,s} \tag{6}$$

where $AM_{a,c,s}$ represents the amenable TB mortality rate and $Population_{a,c,s}$ denotes the population size for the specific stratum.

Table 1
Age and Sex specific global HIV-negative tuberculosis total and amenable mortality rates in 2021 by drug resistance type.

Mortality rate (Per 100,000) (95%UI)										
		Male			Female			Both		
		Total	Amenable	Amenable/ Total (%)	Total	Amenable	Amenable/ Total (%)	Total	Amenable	Amenable/ Total (%)
TB	All	18.32 (16.32-21.94)	8.79 (8.27-9.31)	46.50 (38.87-54.13)	11.12 (10.09-12.27)	7.48 (7.28-7.67)	64.29 (57.78-70.80)	14.74 (13.31-16.65)	8.12 (7.81-8.43)	53.04 (46.70-59.38)
	0-14 years	3.06 (2.21-4.16)	3.01 (2.73-3.29)	98.21 (64.64-99.99)	3.99 (3.16-4.97)	3.87 (3.62-4.11)	96.75 (74.02-100)	3.51 (2.67-4.45)	3.42 (3.19-3.65)	97.38 (71.59-99.99)
	15-49 years	11.21 (9.96-13.19)	7.02 (6.48-7.56)	62.60 (52.40-72.79)	6.34 (5.69-7.11)	5.62 (5.43-5.81)	88.61 (78.24-98.98)	8.80 (7.96-9.81)	6.33 (6.01-6.65)	71.83 (63.50-80.15)
	50-64 years	34.71 (31.31-40.54)	16.70 (14.47-18.93)	48.07 (37.97-58.17)	17.01 (15.51-18.54)	12.81 (12.07-13.56)	75.28 (65.47-85.03)	25.73 (23.72-28.56)	14.60 (13.31-15.89)	56.70 (48.14-65.26)
	65-74 years	59.85 (52.88-72.90)	30.82 (26.20-35.44)	51.44 (39.94-62.95)	29.99 (27.07-33.32)	23.23 (21.69-24.76)	77.39 (67.96-86.81)	44.14 (39.99-50.18)	26.77 (24.17-29.38)	60.60 (51.54-69.66)
DS-TB	All	16.52 (14.02-19.86)	7.92 (7.42-8.43)	46.46 (37.73-55.18)	10.02 (8.71-11.32)	6.71 (6.52-6.91)	64.07 (55.52-72.62)	13.28 (11.55-15.20)	7.30 (6.98-7.61)	52.88 (45.27-60.20)
	0-14 years	2.89 (1.99-3.82)	2.75 (2.49-3.01)	98.21 (64.80-99.99)	3.64 (2.86-4.51)	3.52 (3.29-3.75)	96.68 (73.93-100)	3.21 (2.42-4.06)	3.13 (2.91-3.34)	97.38 (71.59-100)
	15-49 years	10.06 (8.52-11.99)	6.32 (5.80-6.84)	62.75 (50.76-74.74)	5.68 (4.87-6.46)	5.03 (4.84-5.23)	88.55 (75.81-100)	7.90 (6.78-8.98)	5.68 (5.36-6.01)	71.89 (61.13-82.66)
DS-TB	50-64 years	31.22 (26.74-36.69)	15.06 (12.90-17.21)	48.19 (36.78-59.58)	15.31 (13.44-16.92)	11.48 (10.74-12.23)	74.96 (63.06-86.85)	23.15 (20.47-25.99)	13.09 (11.81-14.37)	56.49 (46.49-66.49)
	65-74 years	54.03 (45.56-66.13)	27.61 (23.04-32.18)	51.05 (38.22-63.89)	26.99 (22.89-30.44)	20.77 (19.20-22.34)	76.92 (64.78-89.05)	39.80 (34.17-45.97)	23.96 (21.31-26.61)	60.16 (49.10-71.22)
MDR-TB	All	1.69 (0.67-3.31)	0.79 (0.64-0.93)	45.72 (8.61-82.81)	1.04 (0.40-2.07)	0.70 (0.60-0.80)	64.64 (11.70-99.99)	1.35 (0.53-2.68)	0.74 (0.62-1.86)	52.76 (9.90-95.62)
	0-14 years	0.25 (0.10-0.50)	0.24 (0.18-0.30)	97.95 (14.70-98.57)	0.34 (0.15-0.66)	0.33 (0.26-0.40)	97.07 (21.12-98.18)	0.29 (0.12-0.56)	0.28 (0.22-0.35)	97.44 (21.16-100)
	15-49 years	1.07 (0.43-2.11)	0.65 (0.49-0.81)	61.20 (10.71-71.44)	0.61 (0.23-1.26)	0.55 (0.45-0.64)	88.63 (12.10-95.94)	0.84 (0.33-1.70)	0.60 (0.47-0.73)	71.08 (11.48-84.43)
	50-64 years	3.20 (1.31-6.23)	1.46 (0.89-2.03)	45.57 (4.87-86.26)	1.59 (0.61-3.25)	1.18 (0.82-1.55)	74.34 (5.86-97.20)	2.39 (0.96-4.68)	1.31 (0.87-1.76)	54.92 (6.34-65.83)
	65-74 years	5.40 (2.05-10.75)	2.76 (1.39-4.13)	51.06 (2.72-64.01)	2.82 (1.06-5.89)	2.19 (1.34-3.04)	77.54 (4.62-97.20)	4.04 (1.51-8.16)	2.44 (1.38-3.50)	60.40 (4.21-73.16)
XDR-TB	All	0.13 (0.06-0.25)	0.05 (0.03-0.06)	33.39 (8.19-58.60)	0.07 (0.03-0.13)	0.04 (0.03-0.05)	56.67 (12.56-100)	0.10 (0.04-0.18)	0.04 (0.03-0.05)	41.02 (10.63-71.41)
	0-14 years	0.01 (0-0.02)	0.01 (0-0.02)	96.54 (19.54-100)	0.01 (0-0.03)	0.01 (0-0.02)	94.59 (19.31-100)	0.01 (0-0.03)	0.01 (0-0.02)	95.41 (18.13-100)
	15-49 years	0.08 (0.04-0.15)	0.04 (0.03-0.05)	48.12 (12.47-83.77)	0.04 (0.02-0.08)	0.03 (0.02-0.04)	82.47 (17.66-100)	0.06 (0.03-0.12)	0.04 (0.03-0.05)	59.30 (15.05-71.40)
	50-64 years	0.29 (0.13-0.51)	0.09 (0.05-0.12)	30.29 (6.54-54.03)	0.11 (0.04-0.20)	0.07 (0.05-0.09)	64.94 (10.41-86.93)	0.19 (0.09-0.35)	0.08 (0.05-0.10)	39.30 (8.46-70.14)
	65-74 years	0.43 (0.17-0.83)	0.16 (0.08-0.23)	36.53 (3.20-69.85)	0.20 (0.07-0.36)	0.13 (0.09-0.18)	69.17 (10.72-95.63)	0.30 (0.12-0.59)	0.14 (0.08-0.20)	46.97 (5.35-88.65)

Note: Drug-susceptible Tuberculosis (DS-TB), Multidrug-resistant Tuberculosis (MDR-TB), and Extensively drug-resistant Tuberculosis (XDR-TB).

The VSL represents the monetary amount that individuals are willing to pay for a marginal reduction in mortality risk. We estimated country-specific peak VSL using an income elasticity (IE) approach (Gerstl et al., 2023):

$$VSL_{p,c} = VSL_{p,USA} \times \left(\frac{GDP_{c,per}}{GDP_{USA,per}}\right)^{IE} \tag{7}$$

where $VSL_{p,c}$ and $GDP_{c,per}$ denote the peak VSL and the GDP per capita in c-th country, respectively (Kniesner & Viscusi, 2019). We selected an IE of 1 for the baseline analysis to minimize assumptions about willingness to pay after adjustment for GDP per capita and purchasing power. $VSL_{p,USA}$ is based on the 2021 VSL estimate (US$11.8 million) provided by the U.S. Department of Transportation (UDo, 2024).

Because VSL represents the value of an entire lifetime, it was converted into an annualized measure—the Value of a Statistical Life Year (VSLY)—to match the units of YLL (Nandi et al., 2022). Both VSL and VSLY vary with age and will not remain constant over the lifetime of an individual, It has been observed that VSL fits a polynomial inverted U-shaped curve (Aldy & Viscusi, 2008). An age-specific VSL (VSL_a) can be estimated with the following (Ranganathan et al., 2021):

$$VSL_{a,c,s} = VSL_{p,c}\, f(a) \tag{8}$$

$$f(a) = -0.36456\left(\frac{a}{LE_{c,s}}\right)^4 - 5.65959\left(\frac{a}{LE_{c,s}}\right)^3 + 4.3871\left(\frac{a}{LE_{c,s}}\right)^2 + 1.1505\left(\frac{a}{LE_{c,s}}\right) + 0.0008447 \tag{9}$$

where a represents age, $LE_{c,s}$ is the life expectancy for each country 'c' and sex 's', and $f(a)$ is a polynomial function that adjusts a country's peak VSL to VSL_a based on the proportion of life lived.

Given this age-specific VSL, the VSLY can be derived by equating it to the discounted sum of annual life-years lost (Alkire et al., 2018; Ranganathan et al., 2021)

$$VSL_{a,c,s} = VSLY_{a,c,s}\int_a^{LE_{a,c,s}} e^{-dr(x-a)}dx \tag{10}$$

Substitute equation (8) for $VSL_{a,c,s}$ into (10):

$$VSL_{p,c}\, f(a) = VSLY_{a,c,s}\int_a^{LE_{a,c,s}} e^{-dr(x-a)}dx \tag{11}$$

Integrate and solve for $VSLY_{a,c}$ to arrive at:

$$VSLY_{a,c,s} = VSL_{p,c}\, f(a)\frac{-r}{e^{r\left(a-LE_{a,c,s}\right)}-1} \tag{12}$$

$VSLY_{a,c,s}$ then multiplied by age-sex-country-specific YLLs due to amenable TB mortality to obtain the age-sex-country-specific VLW.

$$VLW_{a,c,s} = VSLY_{a,c,s} \times YLL_{AM(a,c,s)} \tag{13}$$

The total VLW for each country was obtained by summing VLW across all age groups, with results expressed in international dollars (2021 purchasing power parity) (Table S3). We assessed the proportion of VLW due to amenable TB mortality relative to the local GDP for each country (Figs. 2–3) and calculated the economic losses for each income level group (Table S3). At the global level, economic losses were aggregated from 186 countries with available economic data (Table 2). To investigate the association between socioeconomic development and amenable TB mortality, we used Spearman's rank correlation coefficient and linear regression models to explore the relationships between the Socio-Demographic Index (SDI), current health expenditure per capita, and the proportion of amenable TB mortality, as well as the proportion of VLW relative to GDP (Figs. S6 and S9).

To estimate the uncertainty in amenable TB mortality and economic value, we used the Monte Carlo simulation method. Specifically, we sampled 10,000 times from the 95% uncertainty interval (UI) boundaries of total TB mortality and TB mortality attributable to behavioral risks, based on GBD data with gamma distribution. We estimated uncertainty for all model outputs by calculating the 2.5th and 97.5th percentiles and denoting them as 95% UIs. We also did sensitivity analyses to evaluate the impact of income elasticity on VSL valuation by setting an IE of 0.8 for countries with a GDP per capita higher than that of the United States and 1.5 for countries with a lower GDP per capita (Chang et al., 2024; Verguet et al., 2024) (Table S4, Table S5). To assess the consistency of our estimates over time, we applied the same analytical framework to the most recent GBD 2022 and 2023 data. All procedures were identical to those used for the 2021 analysis to ensure comparability across years (Tables S6–S12). All statistical analyses were performed using R 4.4.2.

3. Results

3.1. Amenable TB mortality

Global amenable TB deaths in 2021 were estimated at 616,736 (95% UI: 593,599–639,871), including 554,224 from DS-TB (95% UI: 530,885–577,633), 56,359 from MDR-TB (95% UI: 47,485–65,233), and 3260 from XDR-TB (95% UI: 2733–3746). These accounted for 53.04% (95% UI: 46.70%-59.38%) of the total TB deaths (1,162,796 95% UI: 1,050,008–1,313,985) worldwide (Table 1, Table S1). The global amenable TB mortality rate in 2021 was 8.12 (95% UI: 7.81-8.43) per 100,000. The amenable mortality rates for DS-TB, MDR-TB and XDR-TB were 7.30 (95% UI: 6.98-7.61), 0.74 (95% UI: 0.62-1.86), and 0.04 (95% UI: 0.03-0.05) per 100,000, respectively (Table 1). The highest proportion of amenable TB mortality relative to total TB mortality was 97.38% (95% UI: 71.59%-99.99%) in the 0-14 age group, followed by 71.83% (95% UI: 63.50%-80.15%) in the 15-49 age group. This proportion of amenable TB mortality was higher for females (64.29%) than for males (46.50%). Similar age and sex distributions were observed across different TB drug resistance types (Table 1).

At the regional level (Fig. S1), both the amenable TB mortality rate and the proportion of amenable mortality relative to total TB mortality were highest in Central Sub-Saharan Africa, with 34.71 (95% UI: 30.21-39.20) per 100,000 and 67.18% (95% UI: 53.93%-71.20%), respectively. The amenable TB mortality rate was lowest in Australasia at 0.02 (95% UI: 0.01-0.03) per 100,000, while the proportion was lowest in the High-income Asia Pacific region with 1.48% (95% UI: 0.34%-3.84%). In North

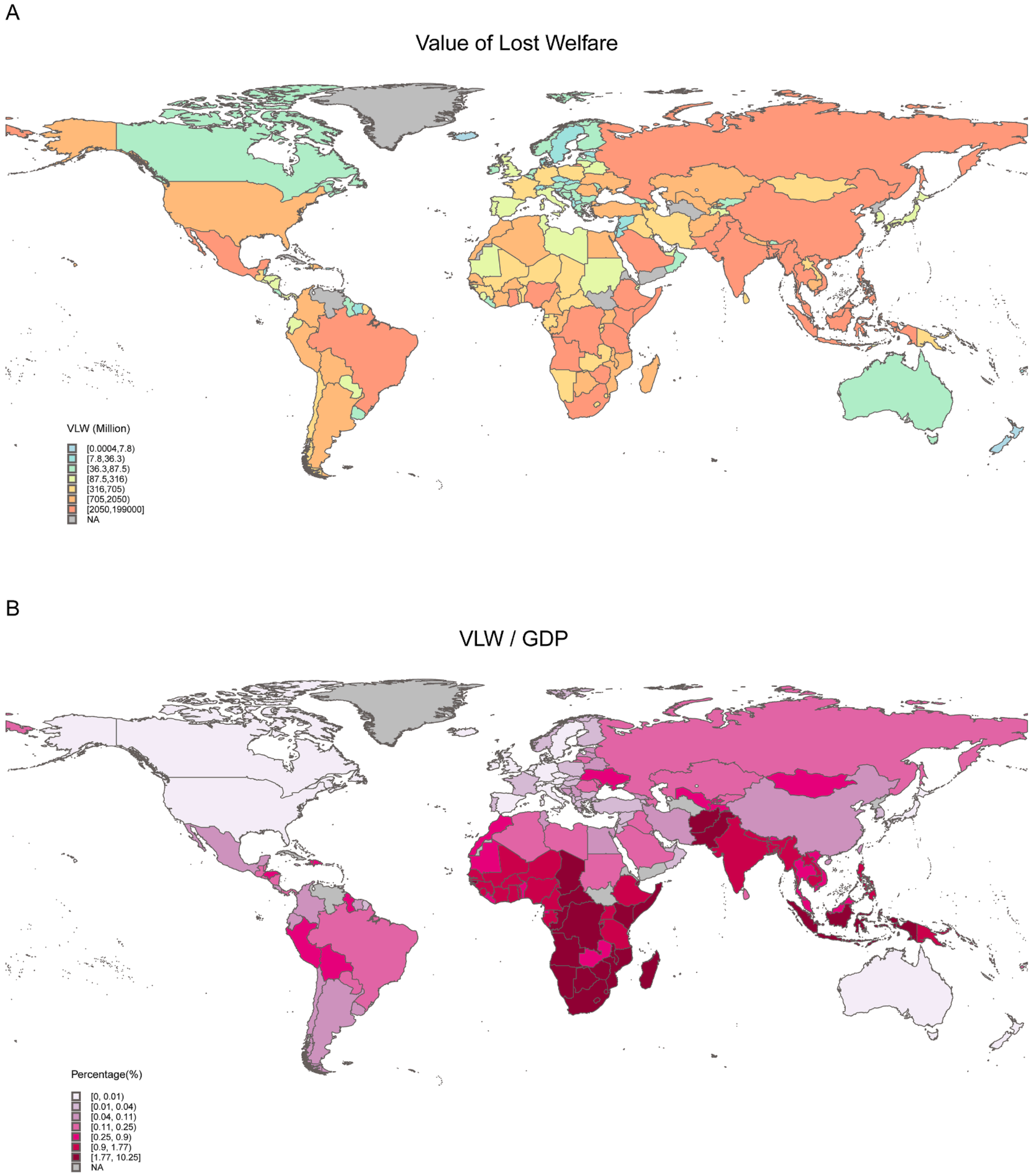

Fig. 2. HIV-negative TB resulting (A)Value of Vost Welfare (VLW), (B) VLW as a percentage of gross domestic product (GDP) for 186 countries and territories in 2021.

Africa and the Middle East, although the amenable TB mortality rate was only 2.07 (95% UI: 1.91-2.23) per 100,000, the proportion of amenable deaths relative to total TB deaths remained high at 60.54%. (Fig. S1). The differences in the amenable TB mortality rate and the proportion of amenable TB mortality to total TB mortality by drug resistance type at the regional level are shown in Fig. S2. By income group, the amenable TB mortality rate and proportion were highest in low-income countries, at 22.07(95% UI: 20.56–23.58) per 100,000 and 65.90% (95% UI: 57.06%–74.53%), respectively (Table S2).

At the country level (Fig. 1), the highest and lowest amenable TB mortality rates were 104.67 (95% UI: 90.60-118.74) per 100,000 in the Central African Republic and 8.16×10^{-5} (95% UI: 0-0.005) per 100,000 in Malta, respectively. The highest and lowest proportions of amenable TB mortality relative to total TB mortality were 81.23% (95% UI: 61.70%-92.53%) in Somalia and 0.04% (95% UI: 0%-0.10%) in Malta. The country-level differences in DS-TB, MDR-TB, and XDR-TB total mortality rates, amenable mortality rates, and the proportion of amenable TB mortality to total TB mortality are shown in Figs. S3–S5.

We found a significant negative association between the proportion of amenable TB mortality to total TB mortality and per capita health expenditure, SDI (Fig. S6). Specifically, in countries with per capita health expenditure below $1,239, the proportion of amenable TB mortality decreased rapidly as health expenditure increased, with a regression coefficient of

Fig. 3. Value of lost welfare (VLW) due to TB amenable mortality as a percentage of gross domestic product (GDP) in 2021 for the global burden of disease (GBD) regions and countries by four age groups.

$\beta = -0.00024$ ($P < 0.001$). This suggested that for every $100 increase in per capita health expenditure, the proportion of amenable TB mortality decreased by 2.4%. In countries with per capita health expenditure above $1,239, the decline was slight, with a regression coefficient of $\beta = -0.00004$ ($P < 0.001$). This indicated that for every $100 increase in per capita health expenditure, the proportion of amenable TB mortality decreased by only 0.4%. Additionally, the proportion of amenable mortality relative to total TB mortality decreased as the SDI increased ($\beta = -0.966$,$P < 0.001$), meaning that for every 0.01 increase in SDI, the proportion of amenable TB mortality decreased by 0.97%.

Table 2
Value of Lost Welfare (VLW) due to global HIV-negative TB amenable mortality in 2021, VLW expressed as equivalent proportion of 2021 GDP by drug resistance type.

		VLW (2021 International Dollars, Billions)			VLW/GDP (%)		
		Male	Female	Both	Male	Female	Both
TB	All	264.39 (244.62-284.16)	215.95 (209.17-222.72)	480.75 (457.11-504.39)	0.170 (0.157-0.183)	0.139 (0.134-0.143)	0.309 (0.293-0.324)
	0-14 years	1.23 (1.15-1.32)	1.34 (1.27-1.41)	2.57 (2.45-2.70)	0.079×10^{-2} (0.074-0.085) $\times 10^{-2}$	0.086×10^{-2} (0.081-0.090) $\times 10^{-2}$	0.002 (0.001-0.002)
	15-49 years	134.38 (121.45-147.32)	99.98 (96.11-103.84)	234.26 (219.72-248.79)	0.086 (0.078-0.095)	0.064 (0.062-0.067)	0.151 (0.141-0.160)
	50-64 years	89.95 (76.42-103.47)	74.79 (69.99-79.58)	164.03 (147.56-180.51)	0.058 (0.049-0.066)	0.048 (0.045-0.051)	0.105 (0.095-0.116)
	65-74 years	38.83 (32.49-45.17)	39.85 (37.03-42.66)	79.89 (71.17-88.60)	0.025 (0.021-0.029)	0.026 (0.024-0.027)	0.051 (0.046-0.060)
DS-TB	All	235.90 (216.72-255.09)	192.31 (185.40-199.22)	427.84 (404.05-451.62)	0.152 (0.139-0.164)	0.124 (0.119-0.128)	0.275 (0.260-0.290)
	0-14 years	1.12 (1.04-1.20)	1.21 (1.14-1.28)	2.33 (2.20-2.46)	0.072×10^{-2} (0.067-0.077) $\times 10^{-2}$	0.078×10^{-2} (0.073-0.082) $\times 10^{-2}$	0.001 (0.001-0.002)
	15-49 years	119.68 (107.15-132.20)	88.97 (85.01-92.93)	208.54 (193.66-223.41)	0.077 (0.067-0.085)	0.057 (0.055-0.060)	0.134 (0.124-0.144)
	50-64 years	80.45 (67.33-93.57)	66.58 (61.74-71.42)	145.69 (129.32-162.06)	0.052 (0.043-0.060)	0.043 (0.040-0.046)	0.094 (0.083-0.104)
	65-74 years	34.66 (28.44-40.88)	35.55 (32.62-38.48)	71.28 (62.52-80.03)	0.022 (0.018-0.026)	0.023 (0.021-0.025)	0.046 (0.040-0.051)
MDR-TB	All	23.79 (19.12-28.47)	20.53 (16.87-24.19)	44.29 (35.83-52.74)	0.015 (0.012-0.018)	0.013 (0.011-0.016)	0.028 (0.023-0.034)
	0-14 years	0.10 (0.07-0.13)	0.12 (0.09-0.16)	0.23 (0.16-0.29)	0.067×10^{-3} (0.048-0.086) $\times 10^{-3}$	0.079×10^{-3} (0.056-0.102) $\times 10^{-3}$	0.015×10^{-2} (0.011-0.019) $\times 10^{-2}$
	15-49 years	12.74 (9.42-16.06)	9.88 (7.73-12.03)	22.64 (16.95-28.32)	0.008 (0.006-0.010)	0.006 (0.005-0.008)	0.015 (0.011-0.018)
	50-64 years	7.65 (4.76-10.55)	6.91 (4.39-9.42)	14.47 (9.10-19.85)	0.005 (0.003-0.007)	0.004 (0.003-0.006)	0.009 (0.006-0.013)
	65-74 years	3.29 (1.73-4.86)	3.62 (2.07-5.18)	6.95 (3.77-10.13)	0.002 (0.001-0.003)	0.002 (0.001-0.003)	0.004 (0.002-0.007)
XDR-TB	All	2.17 (1.85 -2.48)	1.73 (1.51-1.95)	3.87 (3.33-4.41)	0.001 (0.001-0.002)	0.001 (0.001-0.002)	0.002 (0.002-0.003)
	0-14 years	0.007 (0.005-0.009)	0.007 (0.005-0.009)	0.014 (0.011-0.018)	0.046×10^{-4} (0.035-0.057) $\times 10^{-4}$	0.047×10^{-4} (0.035-0.060) $\times 10^{-4}$	0.092×10^{-4} (0.070-0.114) $\times 10^{-4}$
	15-49 years	1.24 (0.99-1.48)	0.86 (0.72-1.00)	2.10 (1.71-2.49)	0.080×10^{-2} (0.064-0.095) $\times 10^{-2}$	0.056×10^{-2} (0.047-0.064) $\times 10^{-2}$	0.001 (0.001-0.002)
	50-64 years	0.64 (0.47-0.82)	0.54 (0.40-0.68)	1.16 (0.84-1.47)	0.041×10^{-2} (0.030-0.053) $\times 10^{-2}$	0.035×10^{-2} (0.026-0.044) $\times 10^{-2}$	0.074×10^{-2} (0.054-0.095) $\times 10^{-2}$
	65-74 years	0.28 (0.19-0.37)	0.32 (0.22-0.41)	0.60 (0.41-0.79)	0.018×10^{-2} (0.012-0.024) $\times 10^{-2}$	0.020×10^{-2} (0.001-0.003) $\times 10^{-2}$	0.038×10^{-2} (0.026-0.051) $\times 10^{-2}$

3.2. Economic losses due to amenable TB mortality

Based on the VLW approach, we estimated that global economic losses due to amenable TB mortality in 2021 were $480.75 (2021 international dollars; 95% UI: 457.11-504.39) billion for 186 countries, equivalent to 0.31% (95% UI: 0.29%-0.32%) of the global GDP (Table 2). Economic losses due to amenable mortality from DS-TB, MDR-TB and XDR-TB were $427.84 (95% UI: 404.05-451.62) billion, $44.29 (95% UI: 35.83-52.74) billion, and $3.87 (95% UI: 3.33-4.41) billion, respectively. These values corresponded to 0.28% (95% UI: 0.26%-0.29%), 0.03% (95% UI: 0.02%-0.03%), and 0.002% (95% UI: 0.002%-0.003%) of the global GDP. Among different age and gender groups, the VLW from TB amenable deaths was highest in the 15-49 age group ($234.26 billion), lowest in the 0-14 age group ($2.57 billion), and higher in males ($264.39 billion) than in females ($215.95 billion) (Table 2). Applying the same analytical framework to GBD 2022 and 2023 data yielded comparable estimates. Global economic losses were $525.60 billion in 2022 and $557.85 billion in 2023, corresponding to 0.32% and 0.33% of global GDP, respectively (Table S6).

The differences in economic losses due to amenable TB mortality at the country level in 2021 are shown in Fig. 2 and Table S3. India had the highest economic loss due to TB at $199,426 (95% UI: $177,957-220,896) million, representing 1.75% (95% UI: 1.56%-1.94%) of its GDP. The country with the highest relative economic losses from amenable TB mortality was Central African Republic (10.25%, 95% UI: 8.64%-11.87%), with an absolute economic loss of $591.62 (95% UI: $498.20-685.03) million. By income group, low-income countries experienced the highest relative economic losses due to amenable TB mortality, amounting to 1.73% (95% UI: 1.65%-1.81%) of their GDP, equivalent to $24,787(95% UI: $23,602 -25,972) million. Lower-middle income countries faced the largest absolute economic losses at $376,767 (95% UI: $353,92-399,642) million, representing 1.30% (95% UI: 1.22%-1.38%) of their GDP (Table S3). The relative economic losses due to amenable mortality from DS-TB, MDR-TB, and XDR-TB at the country level are shown in Fig. S8.

The age distribution of relative economic losses due to amenable TB mortality across different GBD regions and countries is shown in Fig. 3. The values of relative economic losses varied significantly across GBD regions. Among them, the highest relative economic losses due to TB in the Southern Sub-Saharan Africa region were 2.45% (95% UI: 2.17%–2.73%), while the lowest relative economic losses due to TB in the High-income Asia Pacific and Australasia region were 0.006% (95% UI: 0.001%–0.028%). In most regions (17/18), the economic losses caused by amenable TB mortality predominantly came from the 15-49 age group. However, in the Western Europe region, amenable TB mortality mainly contributed to economic losses in the 65-74 age group. Across all regions and countries, the smallest economic losses due to TB occurred in the 0-14 age group (Fig. 3).

There was a negative association between the relative economic losses due to amenable TB mortality and SDI (Fig. S9). In countries with an SDI below 0.72, the relative economic losses decreased more sharply with increasing SDI (β = -0.07, $P < 0.001$). This suggested that for every 0.01 increase in SDI, the relative economic losses decreased by 0.07%. In contrast, in countries with an SDI above 0.72, the relative economic losses decreased more slowly with increasing SDI (β = -0.007, $P < 0.001$), meaning that for every 0.01 increase in SDI, the relative economic losses decreased by 0.007%.

In the sensitivity analysis, country- and region-level estimates of economic losses due to amenable TB mortality in 2021, using income elasticities (IEs) of 0.8 and 1.5, are presented in Tables S4 and S5. India still experienced the highest absolute economic loss from amenable TB mortality, estimated at $68,613 (95% UI: $61,227-76,000) million, which accounted for 0.60% (95% UI: 0.53-0.66) of its GDP. Eswatini had the highest relative economic loss, with amenable TB mortality accounting for 1.55% (95% UI:1.27%-1.83%) of its GDP, despite an absolute economic loss of $188.16 (95% UI: $154.18-222.14) million. Overall, although the VLW estimates differed from those in the main analysis due to variations in IE, the ranking of economic losses from amenable TB mortality across countries remained similar.

4. Discussion

Our study estimated amenable TB mortality and its economic burden in 2021. We found that the global amenable TB mortality across 203 countries was 616,736, with 53.04% of TB-related deaths potentially avertable through access to high-quality healthcare. The total economic losses from amenable TB mortality in 186 countries amounted to $480.75 billion, equivalent to approximately 0.31% of global GDP in 2021. Importantly, applying the same framework to GBD 2022 and 2023 data yielded comparable proportions of GDP loss (approximately 0.32% and 0.33%, respectively), suggesting that the magnitude of these economic losses has remained relatively stable in recent years (Table S6). The distribution of amenable TB mortality and its economic losses was highly uneven across regions (Figs. 1–2). The geographical patterns of amenable TB mortality and its proportion closely aligned with the overall TB mortality burden (Zhang et al., 2024). The highest absolute economic losses from amenable TB mortality occurred in the South Asia region and lower-middle-income countries, highlighting the challenges of TB control in resource-limited settings. To the best of our knowledge, this study is the first to estimate the global, regional, and country-level economic value of reducing amenable TB mortality.

Our study shows that 53.04% of global total TB deaths in 2021 were amenable to healthcare, which is close to previous estimate of 55% for low- and middle-income countries (LMICs) in 2016 (Kruk et al., 2018). Despite the decline in total TB mortality over the past years (WHO, 2024), the proportion of amenable deaths remains above 50%. Notably, we found that 97.38% of TB-related deaths among children aged 0–14 years were amenable to prevention with adequate healthcare services (Table 1). This aligns with a previous study that over 96% of pediatric TB fatalities occurred in children aged 0-14 years who did not receive anti-TB treatment (Dodd et al., 2017). These results highlight the urgent need to enhance access to timely TB diagnosis and treatment for children. Moreover, we found that the majority of amenable TB mortality (90.10%) occurred in DS-TB (Table 1). This is primarily because DS-TB accounts for the largest proportion (94.46%) of the overall TB burden, as most newly diagnosed TB cases are sensitive to first-line drugs (Collaborators, 2024a; Furin et al., 2019). Providing high-quality healthcare services to newly diagnosed TB patients can significantly reduce the risk of drug resistance and recurrence while lowering the overall TB mortality burden. Previous studies have found that most TB deaths occur within the first eight weeks of treatment, largely due to extensive disease, comorbidities, and moderate to severe undernutrition (Bhargava & Bhargava, 2020; Birlie et al., 2015; Waitt et al., 2011). Therefore, in addition to uninterrupted TB drug therapy (Wolde et al., 2025), healthcare providers should conduct comprehensive individual risk assessments at treatment initiation, perform implement continuous monitoring during the treatment process, and establish specific hospitalization and referral protocols to ensure integrated care (Bhargava & Bhargava, 2020; Bhargava et al., 2024). Furthermore, we found that countries with lower per capita healthcare expenditure and SDI tended to have higher proportions of amenable TB deaths (Fig. S6). This is because TB and its adverse outcomes are more prevalent among individuals with lower socioeconomic status, limited education, unemployment, and overcrowded living conditions (Basnyat et al., 2018; Chen et al., 2023; Collaborators, 2024a). These findings emphasize the critical role of socioeconomic development in preventing TB mortality.

Our study estimated that the global economic losses due to amenable TB mortality in 2021 accounted for 0.31% of the global GDP. Ranganathan et al. estimated that in 2015 (with an IE value of 1.5), the VLW due to TB in South Asia and Southern Sub-Saharan Africa amounted to 0.20% and 0.25% of their regional GDP, respectively (Ranganathan et al., 2021), which are similar to our estimates of 0.57% and 1.06% of their regional GDP in 2021 in sensitivity analysis (Table S5). Silva et al. (Silva et al., 2021) found that the economic losses due to TB mortality in 2018 was $151 billion for India and $0.68 billion for the

Central African Republic, which was close to our estimates of $199 billion for India and $0.59 billion for the Central African Republic in 2021. The variations between these results may be attributed to differences in TB mortality burden, economic indicators across years, and research methodologies. These estimated economic losses indicated the substantial and persistent economic burden of TB over time. However, the annual investment in TB prevention, diagnosis, and treatment in 132 low- and middle-income countries was just $5.7 billion (WHO, 2024), and the global TB research funding was only $1.2 billion (Partnership. TST, 2024), which were both far below the funding targets set by the UN High-Level Meeting on TB, calling for $22 billion annually for TB control and $5 billion for TB research (WHO, 2024). This substantial funding gap continues to hinder progress in TB control. Given the practical feasibility of improving the quality of TB healthcare services (Reid et al., 2019; Uplekar et al., 2015) and its huge economic benefits, our findings further emphasize the urgent need for global commitment to fulfilling TB funding pledges.

Our study found that, globally and across most regions, amenable TB mortality results in the greatest economic loss among individuals aged 15–49 (Table 2, Fig. 3). This is primarily because TB mortality is highest in this age group (Collaborators, 2024a), and younger individuals have longer remaining life expectancy in the VLW model, where VSL reaches a peak in middle age due to higher economic productivity (Aldy & Viscusi, 2008; Alkire et al., 2018). Consequently, premature TB deaths among working-age adults lead to significant economic losses from reduced labor supply and household income. However, in Western Europe, the economic loss due to amenable TB deaths is greatest in the 65-74 age group (Fig. 3). This may be because TB burden is generally lower in countries in the Western Europe region, with mortality primarily concentrated among older adults (Collaborators, 2024a). In these countries, TB infection among younger individuals is rare, as most cases come from past latent tuberculosis infection (LTBI) rather than recent local transmission (Abubakar et al., 2012). Additionally, high-quality healthcare services further reduce TB mortality risk in younger populations. In contrast, older adults face a higher risk due to LTBI reactivation as immune function declines (Lönnroth et al., 2015). Age-related deterioration in liver and kidney function can also impair drug metabolism, and the higher prevalence of comorbidities further increases the mortality risk, making this age group the primary contributor to TB deaths (Falzon & van Cauteren, 2008; Hase et al., 2021). In the Western Europe region, the higher TB mortality among older adults offsets the differences in the VSL and YLL between middle-aged and elderly age groups. As a result, the economic loss from amenable TB mortality in Western Europe is predominantly driven by older adults. This finding highlights the need to optimize TB control strategies based on the epidemiological characteristics of different regions and populations to minimize its social and economic impact.

Taken together, these findings suggest substantial scope to reduce TB deaths, but the interpretation of this scope depends on what our benchmark represents. We define amenable TB mortality relative to the lowest case-fatality ratio observed across GBD regions, which should be viewed as a best-observed reference rather than a target achievable through clinical care alone (Kruk et al., 2018). Socioeconomic conditions, including undernutrition, crowded housing, and limited financial protection, can influence care seeking, disease severity at presentation, and treatment completion, and therefore contribute to observed differences in case fatality (Bhargava et al., 2023; Muniyandi, 2023; Wingfield et al., 2014). At the same time, case fatality among diagnosed patients remains closely linked to the quality and timeliness of TB services, including early detection, prompt treatment initiation, adherence support, and management of complications (Bea et al., 2023; Wolde et al., 2025). In line with the WHO End TB Strategy, strengthening TB care also increasingly encompasses patient-centred approaches and social protection measures, which affect access to services and outcomes (Oh et al., 2018; Uplekar et al., 2015). Accordingly, our estimates should be interpreted as the potential welfare gains associated with narrowing the gap between current outcomes and best-observed levels under a comprehensive strengthening approach, rather than as gains achievable through health-sector actions alone.

The study has several limitations. First, we did not include HIV-positive TB patients as we focused on TB-related deaths attributable to healthcare services. Their diagnosis and treatment are more complex, and amenable mortality in this group depends not only on TB care but also on the quality of HIV treatment services and the local HIV burden (Tornheim & Dooley, 2018). Second, our main analysis relied on GBD 2021 estimates, and limitations related to data quality and modelling assumptions in the GBD framework also apply (GDaI, 2024). We additionally analyzed GBD 2022 and 2023 data to assess temporal consistency, but these analyses do not remove the underlying data constraints. We also did not explicitly model COVID-19–related disruptions to TB services, which may affect the accuracy of country-specific estimates during this period. Third, our analysis, based on the VLW framework, focuses primarily on welfare losses associated with mortality and does not incorporate TB morbidity (the non-fatal component of DALYs). Given that the VLW approach is grounded in VSL and YLL, integrating non-fatal health losses into this valuation framework in an internationally comparable manner presents methodological challenges and was therefore beyond the scope of this study. While prior study suggests that YLL accounts for a substantial share of TB DALYs in many settings (Laxminarayan, 2007), illness episodes and post-TB sequelae may still contribute significant health and economic impacts (Lin et al., 2024). Consequently, our estimates may underestimate the total welfare losses and should be interpreted as a conservative lower bound. Finally, the amenable deaths may be overestimated due to underestimated preventable deaths, as the included attributable risk burden may not account for all potential risk factors when calculating preventable deaths. Despite these limitations, our study used data from multiple sources and adopted previously well-validated methodology to estimate the economic burden of TB globally.

5. Conclusion

Our study comprehensively estimates of amenable TB mortality and macroeconomic losses at global, regional, and national levels. The findings reveal that inadequate access to high-quality healthcare is a major contributor to substantial TB mortality and uneven macroeconomic burdens, especially among individuals aged 15-49 and in lower-middle income countries. More TB funding to increase health and economic benefits through improved healthcare access and quality are needed.

CRediT authorship contribution statement

Yueting Liu: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Ni Zhu:** Writing – review & editing, Validation, Methodology, Conceptualization. **Mingwang Shen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Ethics approval and consent to participate

The protocol of the GBD 2021 has been approved by the research ethics board at the University of Washington. The GBD 2021 shall be conducted in full compliance with University of Washington policies and procedures, as well as applicable federal, state, and local laws.

Consent for publication

Not applicable.

Availability of data and materials

The datasets analyzed during the current study are available at https://vizhub.healthdata.org/gbd-results/

Code availability

The source code for the modelling and visualization required to reproduce the results, tables, and figures presented in this study is publicly available on GitHub at https://github.com/Liu-Yueting/Global-Amenable-TB-Economic-Burden.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.idm.2026.04.003.

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